
風扇簡介

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風扇簡介

風扇

風扇與壓縮機之工作氣體壓力區分：

$\Delta P < 0.1 \text{ kg/cm}^2 \rightarrow$ 風扇

$0.1 \text{ kg/cm}^2 \leq \Delta P \leq 1.0 \text{ kg/cm}^2 \rightarrow$ 鼓風機

$\Delta P > 1.0 \text{ kg/cm}^2 \rightarrow$ 壓縮機

- 風扇構造較為簡單，氣密性較低
- 風扇所輸送之氣體較不具危險性
- 壓縮機所輸送之氣體種類較為廣泛



風扇簡介

風扇操作原理

利用轉子旋轉作功，來推動氣體以達成輸送之功能。

風扇簡介

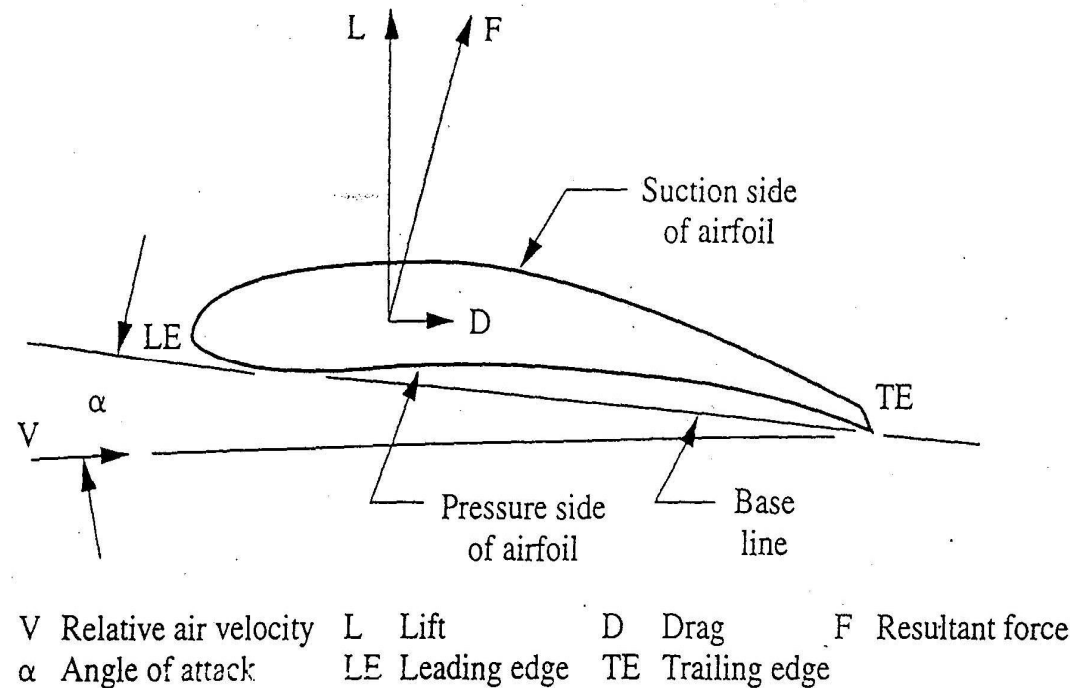
離心式風扇

較之於軸流式風扇，離心式風扇在具備相同葉輪直徑與轉速之條件下，能產生較高之靜壓值；因此一般在工業用途所使用的風扇，以離心式風扇之設置較為廣泛。

風扇簡介

葉片-翼面形葉片

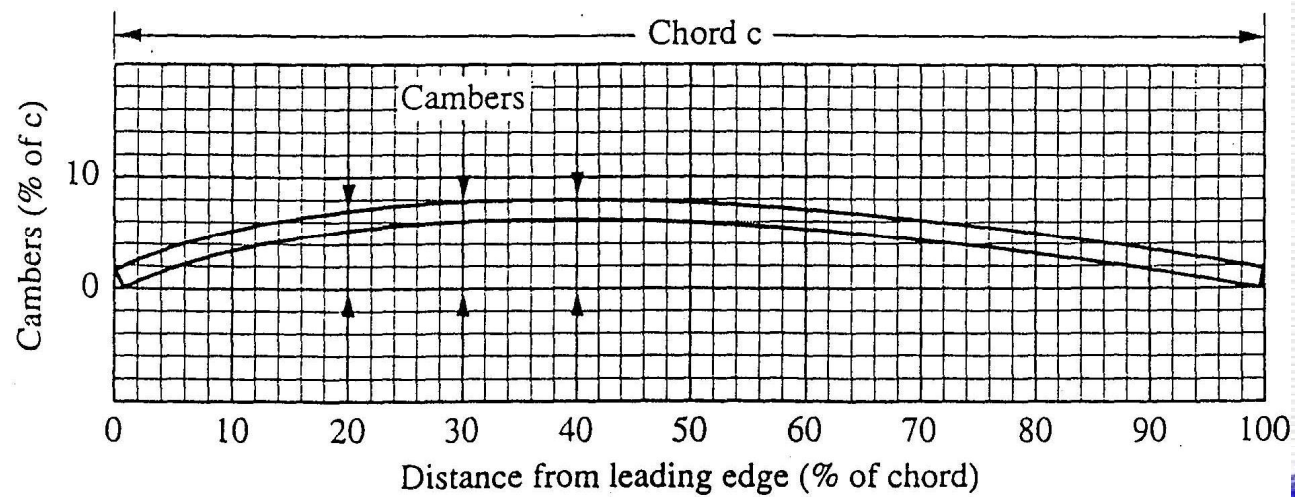
- 效率較佳
- 可調角度範圍大
- 製造較困難
- 製造成本較高



風扇簡介

葉片-等厚厚板葉片

- 可調角度範圍較小
- 製造容易
- 成本較低



風扇簡介

離心式風扇

常用葉片型式分類如下：

- (1) 徑向葉端(Radial Tip; RT)
- (2) 向前曲線(Forward Curved; FC)
- (3) 徑向葉片(Radial Blade; RB)
- (4) 翼截式葉片(Air Foil Blade; AF)
- (5) 向後曲線(Backward Curved; BC)
- (6) 向後傾斜(Backward Inclined; BI)



風扇簡介

離心式風扇

上述葉片型式形狀簡圖如下：



RT



FC



RB



AF



BC



BI

風扇簡介

離心式風扇

徑向葉端葉片(Radial Tip; RT)

- 適用於氣流中夾帶固體粒之工作環境，如氣流中含有中等濃度飛灰、小尺寸顆粒等，
- 適用於具備高濕度之氣流

徑向葉片(Radial Blade; RB)

- 適用於高靜壓、中低流量之應用
- 適用於氣流中含有較大顆粒雜質如粉塵、木屑或金屬碎片等之工作環境



風扇簡介

離心式風扇

前曲式葉片(Forward Curved; FC)

- 適用於低靜壓、中低流量之應用
- 通常使用FC葉片之風扇體積較小
- 不適合流量變化操作
- 可利用之操作範圍較小
- 馬達容易因風扇操作產生過載而損壞
- 僅適用於潔淨之氣體



風扇簡介

離心式風扇

翼面式離心式風扇(Air Foil Blade; AF)

- 效率高
- 噪音較低
- 操作轉速高，可提供較高之靜壓
- 特性曲線表現穩定
- 僅適用於潔淨之氣體

風扇簡介

離心式風扇

向後曲線葉片(Backward Curved; BC)及
向後傾斜葉片(Backward Inclined; BI)

- 設備成本較低
- 適用於溫度變化的操作狀況
- 適用於少量灰塵的操作環境

風扇簡介

離心式風扇與驅動機之排列關係

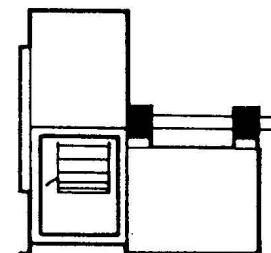
SW - Single Width **DW** - Double Width
SI - Single Inlet **DI** - Double Inlet

Arrangements 1, 3, 7 and 8 are also available with bearings mounted on pedestals or base set independent of the fan housing.

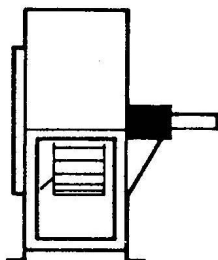
For designation of rotation and discharge, see 99-2406.

For motor position, belt or chain drive, see 99-2407.

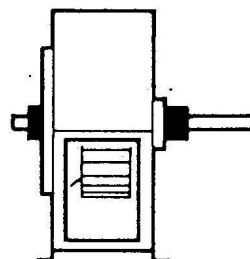
For designation of position of inlet boxes, see 99-2405.



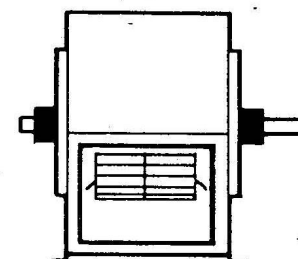
ARR. 1 SWSI For belt drive or direct connection. Impeller overhung. Two bearings on base.



ARR. 2 SWSI For belt drive or direct connection. Impeller overhung. Bearings in bracket supported by fan housing.



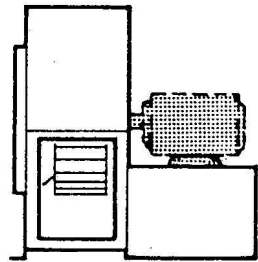
ARR. 3 SWSI For belt drive or direct connection. One bearing on each side and supported by fan housing.



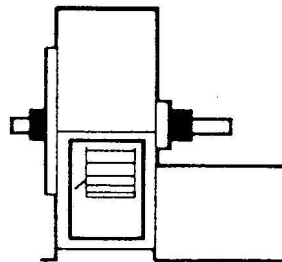
ARR. 3 DWDI For belt drive or direct connection. One bearing on each side and supported by fan housing.

風扇簡介

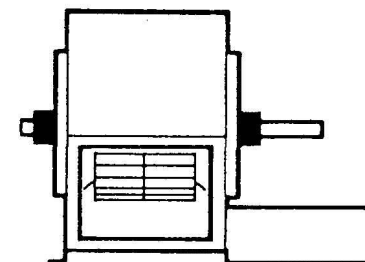
離心式風扇與驅動機之排列關係



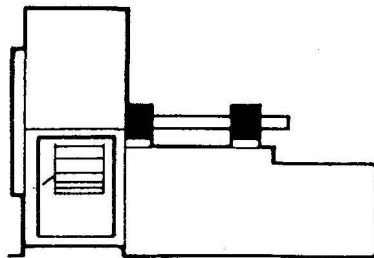
ARR. 4 SWSI For direct drive. Impeller overhung on prime mover shaft. No bearings on fan. Prime mover base mounted or integrally directly connected.



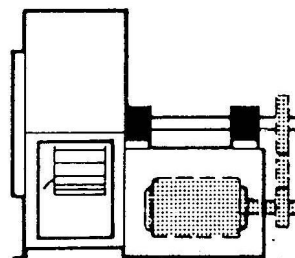
ARR. 7 SWSI For belt drive or direct connection. Arrangement 3 plus base for prime mover.



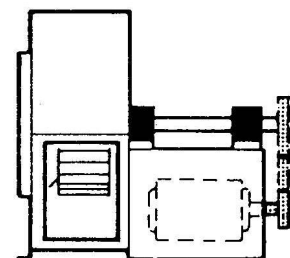
ARR. 7 DWDI For belt drive or direct connection. Arrangement 3 plus base for prime mover.



ARR. 8 SWSI For belt drive or direct connection. Arrangement 1 plus extended base for prime



ARR. 9 SWSI For belt drive. Impeller overhung, two bearings, with prime mover outside base.

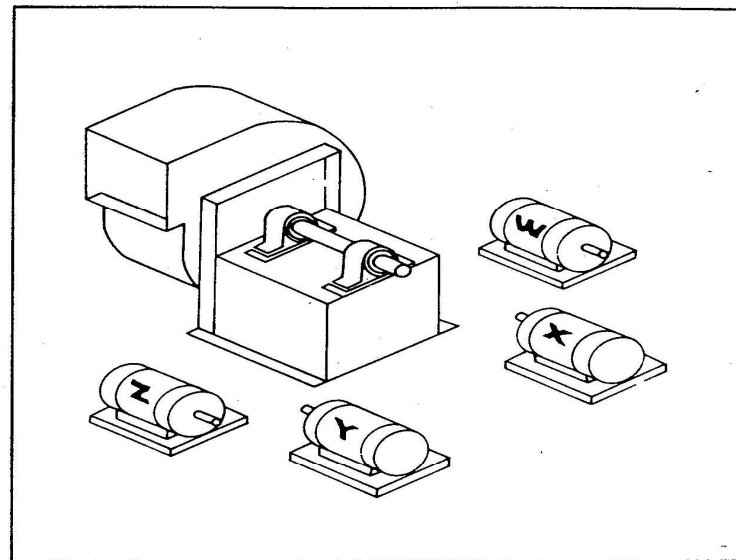


ARR. 10 SWSI For belt drive. Impeller overhung, two bearings, with prime mover inside base.



風扇簡介

皮帶輪驅動風扇與驅動機之排列關係



Location of motor is determined by facing the drive side of fan and designating the motor positions by letters W, X, Y, or Z as the case may be.

2014年



風扇簡介

風扇進氣/排氣口與驅動機之排列關係

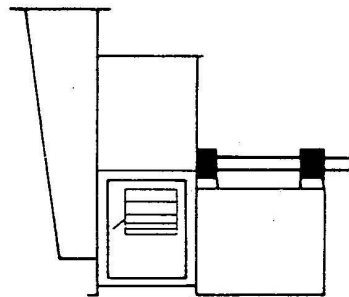
SW - Single Width DW - Double Width
SI - Single Inlet DI - Double Inlet

For designation of rotation and discharge,
For motor position, belt or chain drive,
For designation of position of inlet boxes,

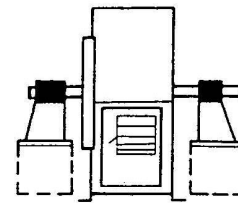
see 99-2406.

see 99-2407.

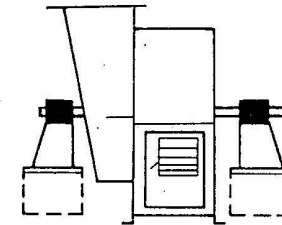
see 99-2405.



ARR. 1 SWSI WITH INLET BOX For belt drive or direct connection. Impeller overhung, two bearings on base. Inlet box may be self-supporting.



ARR. 3 SWSI WITH INDEPENDENT PEDESTAL For belt drive or direct connection fan. Housing is self-supporting. One bearing on each side supported by independent pedestals.

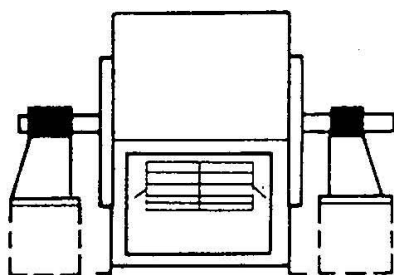


ARR. 3 SWSI WITH INLET BOX AND INDEPENDENT PEDESTALS For belt drive or direct connection fan. Housing is self-supporting. One bearing on each side supported by independent pedestals with shaft extending through inlet box.

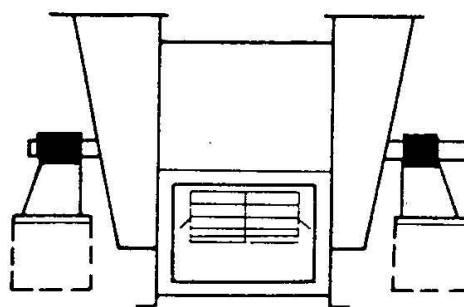


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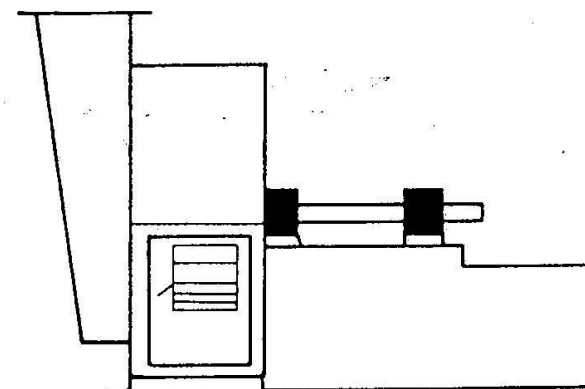
風扇進氣/排氣口與驅動機之排列關係



ARR. 3 DWDI WITH INDEPENDENT PEDESTAL For belt drive or direct connection fan. Housing is self-supporting. One bearing on each side supported by independent pedestals.



ARR. 3 DWDI WITH INLET BOX AND INDEPENDENT PEDESTALS For belt drive or direct connection fan. Housing is self-supporting. One bearing on each side supported by independent pedestals with shaft extending through inlet box.



ARR. 8 SWSI WITH INLET BOX For belt drive or direct connection. Impeller overhung, two bearings on base plus extended base for prime mover. Inlet box may be self-supporting.

敬請指教
謝謝！